

Homework 2

Solutions must be **typed** and **submitted electronically**.

1. Prove that if an encryption scheme Σ has $|\Sigma.\mathcal{K}| < |\Sigma.\mathcal{M}|$ then it cannot satisfy one-time secrecy. Try to structure your proof as an explicit attack on such a scheme (*i.e.*, a distinguisher against the appropriate libraries).

In one-time pad, Enc is a deterministic function but **for full credit**, you should prove the statement even if Enc is randomized. However, you may assume that Dec is deterministic.

Hint: The definition of interchangeability doesn't care about the running time of the distinguisher/calling program. So even an exhaustive brute-force attack would be valid.

2. Let Σ denote an encryption scheme and define Σ^2 to be the following **encrypt-twice** scheme:

$\mathcal{K} = (\Sigma.\mathcal{K})^2$		
$\mathcal{M} = \Sigma.\mathcal{M}$		
$C = \Sigma.C$		
KeyGen:	Enc($((k_1, k_2), m)$):	Dec($((k_1, k_2), (c_1, c_2))$):
$k_1 \leftarrow \Sigma.\mathcal{K}$	$c_1 := \Sigma.\text{Enc}(k_1, m)$	$m_1 := \Sigma.\text{Dec}(k_1, c_1)$
$k_2 \leftarrow \Sigma.\mathcal{K}$	$c_2 := \Sigma.\text{Enc}(k_2, m)$	$m_2 := \Sigma.\text{Dec}(k_2, c_2)$
return (k_1, k_2)	return (c_1, c_2)	if $m_1 \neq m_2$ return err
		return m_1

Prove that if Σ satisfies one-time secrecy, then so does Σ^2 .